# Robotics and Automation

## *Definition of Robotics*

The history of robotics is one that is highlighted by a fantasy world that has provided the inspiration to convert fantasy into reality. Surprisingly, the definition of "robotics" remains controversial between different roboticists. At one end of the spectrum is the science fiction version of a robot, typically one of a human form—an android or humanoid comprising of super features. At the other end of the spectrum is the repetitive, efficient robot of industrial automation which could perform the same task repeatedly with more efficiency and precision than humans. In **ISO 8373, the International Organization for Standardization** defines a robot as "an automatically controlled, reprogrammable, multipurpose manipulator with three or more axes." The definition highlights the functioning of robots that are controlled automatically and could be reprogrammed according to the functionality required by the robots. Moreover, robots are also used as a multipurpose manipulator which can move heavier objects from one place to another. Another definition is given by **"The Robot Institute of America"** which designates a robot as "a reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through various programmed motions for the performance of a variety of tasks." A more inspiring definition is offered by **Merriam Webster**, stating that a robot is "a machine that looks like a human being and performs various complex acts of a human being." In this definition, the robot is described as a simple machine which can perform some complex human actions like walking, talking etc.

## *Modern History (1900 – till now)*

First, **L. Frank Baum's Oz books** imagined and gave ideas about mechanical people like Tin Woodman and Tik-Tok. In real life, **Leonard Torres Quevedo** built the radio system known as **"Telekino"** that could steer a boat from the shore, almost 2 km away, without anyone controlling it. In 1912, **Torres Quevedo** also built a machine that played a **chess game** on its own, with no human inside. He also wrote details about machines that use sensors to sense the world and change their behavior according to the world. The first time, the word "robot" was called in a "**Czech play"** called **"R.U.R"** (1920). In 1930, the first designs for industrial robotic arms appeared. At the 1939 World's fair, a tall robot named **"Elektro"** who could walk, talk and even smoke was introduced. Scientists then started working on the reactions of animal which animals use to control their bodies, which led to a filed called "Cybernetics". **William Grey Walter** built a tortoise robot that could find their way to charging station by

following light. In the 1950s George Devol invented Unimate, the first programmable robot arm, which started factory work in 1961. Japan led the way with humanoid robots like WABOT and later Honda's ASIMO, while arms such as SCARA became more precise and laser-guided weapons showed machines could steer themselves. In the 1990s and 2000s, robots helped in surgery, explored Mars, and entered homes through Sony's AIBO and the Roomba, even as early self-driving cars failed their first big test. By the 2010s, cheap chips from smartphones, plus AI, drones, and self-driving features, made robots small, smart, and everywhere. This brought worries about privacy, safety, and lost jobs, along with surprises like Sophia receiving Saudi citizenship in 2017 and tiny nanobots in 2019 that may one day work inside the human body.

# *USES OF ROBOTS AND AUTOMATION*

## *Robotics in Industrial Manufacturing*

*Robots are machines which are frequently used in modern manufacturing. In most of the industrial systems, robots are frequently used in* ***automating repetitive tasks*** *in an industry. Robots are used for jobs such as welding, packing, assembling etc. This will create room for humans to perform other complex tasks which include making complex ideas, generating strategies and all the tasks which require mental processing because robots are only programed machines which only perform those tasks for which they are being programed and instructed. Robots are mostly used to perform repetitive tasks in industry in which humans can lag and become tired. Robots perform the same tasks repeatedly with same precision and accuracy. Moreover, Robots can work much faster than humans as they don't get tired and remains consistent in their work. Humans cannot work 24/7 (all day) whereas robots can perform actions without being tired and exhausted. It is very efficient and can be used in such dangerous situations where humans cannot work safely.*

### *APPLICATIONS IN INDUSTRIES*

- ***Assembly Operations***
  *Robots are frequently accessed in the manufacturing systems as an assembler of different parts of the products such as automobiles, electronics*

*and appliances. They can do such type of assembly operations much faster and precisely than human beings.*

- ***Welding Purpose***
  *Robots are also frequently accessed by the manufacturers for welding purposes, which are widely used in car body manufacturing, as robots perform precise welding as compared to human beings in a less amount of time. This is very common in automobile industry.*

- ***Material Handling***
  *Robots are also used for the purpose of moving raw material, components and finished products within the factory. They reduce the hard labor and efforts of humans and increase efficiency. For the material handling. Different robots are made such as "robotic arms", "Automated Guided Vehicles (AGVs)" and "Autonomous Mobile Robots (AMRs)".*

- ***Cutting and Machining***
  *Robots are also being used for different machinery purposes such as cutting, drilling, grinding, polishing, and milling operations etc. All these processes are being commanded by humans and robots perform all these operations with perfect accuracy.*

- ***Quality Inspection***
  *Robots are widely used for quality inspection in manufacturing and industrial processes. Here comes the use of the sensors and cameras which are fitted in the robots which examine all products and give the result whether they are according to the required standards of the product or not. Sensors play a vital role in* ***quality inspection*** *by detecting defects, measuring product characteristics, and ensuring that products meet specified standards. They collect data about a product and help manufacturers identify faulty items before they reach customers. These sensors are then put into robots which help in the detection of the quality of the product.*

## *ROBOTS IN MILITIARY:*

*In military as a field, it is very difficult for alone soldiers to complete all their tasks. As their tasks are very difficult, such as walking through the minefields, deactivating unexploded bombs, clearing out hostile buildings etc. For all these tasks robots are used who can do these tasks in a more sophisticated way. But another point is that alone robots cannot do all*

*these works as computer brains are not very sophisticated in terms of ARTIFICIAL INTELLIGENCE. So, for this, robots used in military are remote controlled and are used by humans. Some of its applications are:*

## *APPLICATIONS IN MILITARY*

- ***Bomb disposal**:*

  *It is one of the most important uses of robots in this field as it is done in a safe way and humans stay away from dangerous explosives. It works in such a way that the robots are sent near suspicious objects. The robot is controlled by a trained officer. And cameras and sensors are attached to it from which the officer is continuously monitoring it remotely. If an officer identifies an object as suspicious or potentially dangerous, the robot can activate its built-in safety mechanisms and take appropriate precautionary measures to handle the situation safely and minimize the risk to personnel.*

- ***Search and rescue***

  *The robots are trained for the purpose of search and rescue. Suppose a situation of war is created, and the soldiers are injured but their fellows cannot go there to rescue them. In such type of scenario, robots are used. A robot can enter at any narrow or even dangerous place. The special sensors and cameras attached to them help the rescue teams to locate the people and give them first aid.*

- ***Logistics:***

  *These specially designed robots are used to transfer weapons, equipment, food, ammunition, medicines and various necessary things for soldiers or people, from one place to another. Even at those places where it is impossible for humans to go. They simply reduce human work and do the same work in more precise way.*

- ***Evidence inspection:***

  *Robots can also help the officers to inspect the places where it is difficult to go and find out the evidence. Robots can be sent to any*

*disastrous place, confined space and through the cameras and sensors, record the current situation and provide evidence to officers.*

- ***Underwater crime investigation:***

  *If the investigators need to examine objects underlying the river, then there can be many challenges faced by the inspectors such as the poor visibility, deep water etc. In this case the trained robots are sent that map out the water depth and the object and they create a digital record of the scene. Instead of immediately sending divers into an unfamiliar underwater environment, investigators can first use a remotely operated robot. This gives investigators a clear picture of the scene and a record is created that can be reviewed later and it helps in making further plans.*

## *Use of Space Robots*

- ***Probes & Orbiters:***
  *Deep-space devices that orbit celestial bodies to collect optical, radar, and atmospheric data (e.g., James Webb Telescope, Cassini).*

- ***Rovers & Landers****:*
  *Robotic vehicles that land on planetary surfaces to navigate terrain, drill rocks, analyze soil, and capture HD images (e.g., Curiosity, Perseverance).*

## *Landing & Mobility Technologies*

- ***Aeroshell & Heat Shield****:*
  *Protects the robot from extreme friction and heat during high-speed atmospheric entry.*

- ***Sky Crane****:*
  *Rocket-powered landing system that gently lowers heavy rovers onto foreign surfaces.*

- ***Mobility Systems***:
  *Specialized wheels and suspension designed to traverse rocky terrain, slopes, and low-gravity environments.*

## *Core Operational Mechanics*

***Power Systems:***

- ***Solar Arrays***:
  *Used for inner-planet missions with sufficient sunlight.*

- ***RTGs (Nuclear Power):***
  *Uses decaying Plutonium-238 heat for power in deep-space missions where sunlight is unavailable.*

## *Communication & Autonomy:*

***Signal Delays:***
*Radio signals take 3 to 22 minutes between Earth and Mars, making real-time driving impossible.*

***Autonomous Navigation:***
*Spacecraft executes pre-programmed commands and makes `onboard decisions without direct manual control.*

## *ROBOTICS IN RESEARCH LABORATORIES*

*Laboratory robots are automated machines which are made to perform scientific tasks such as sample handling, automated experiments, collect data, maintaining a certain temperature in laboratory and to do repetitive tasks with accuracy.*

## *FEATURES*

- ***SAFETY:***

*Firstly, it provides safety to the researchers from hazardous and harmful chemicals. Robots handle highly toxic chemicals or radioactive isotopes inside sealed enclosures, keeping personnel safe out of harm way.*

- ***PRECISION AND CONSISTENCY:***

*To be very honest, even the best scientists get tired. After pipetting liquids into a 96 well-plate for three hours, your hands shake a little, your angle shifts, you might lose your focus. For this, robots are made to help researchers consistently. Most of the robots work 24/7 to handle all the projects with precision.*

- ***SMART AUTOMATION:***

*Right now, we have passed the robotics which are not that efficient. Now robotics are linked to AI (Artificial Intelligence). Robots analyze the problems and then send them to AI which then gives the solution to it.*

- ***CLOSED LOOP SYSTEMS****:*

*Traditional labs require that humans have to plan what to do next after completing the first task. But now according to closed loop, an AI algorithm analyzes experimental results instantly and directly instructs the robots that which task should be performed next.*

- ***ACCELERATED DISCOVERY:***

*ACCELRATED DISCOVERY breaks that bottleneck by replacing human wait time with a non-stop feedback loop.*
*Robots do not sleep, do not take breaks and do not eat but then cannot replace the humas as well.*